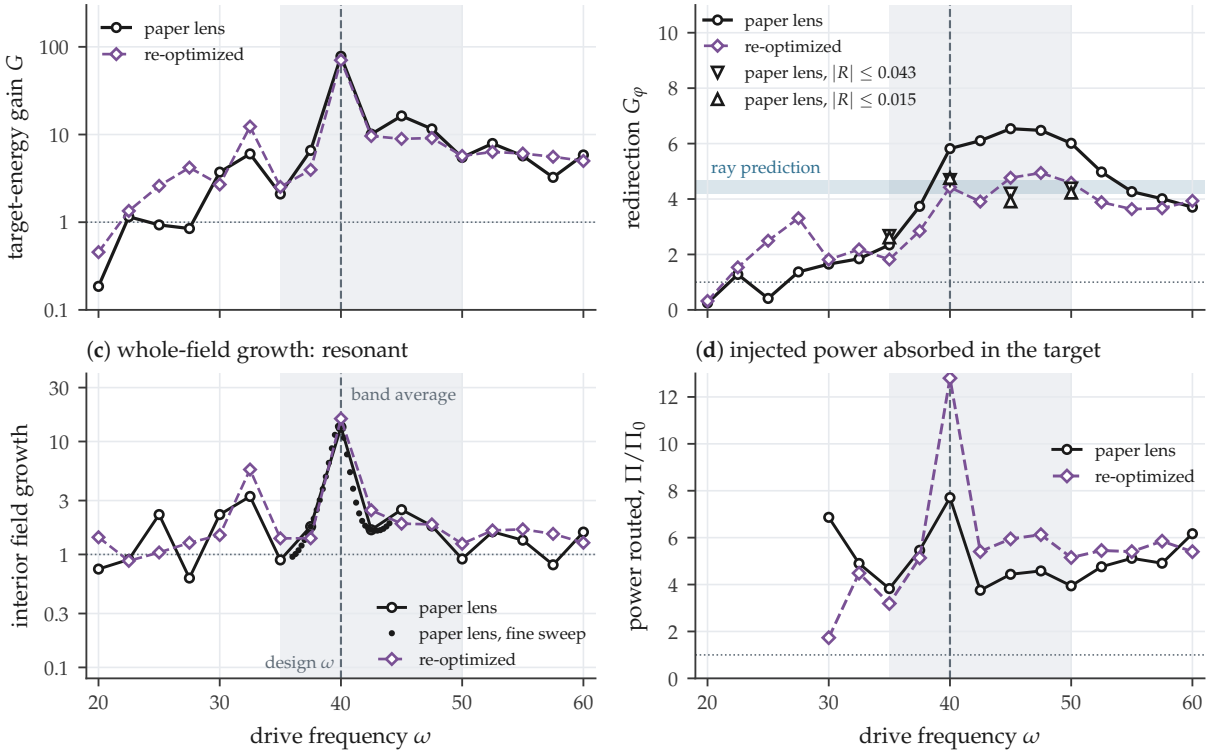

(a) single-frequency gain: narrow peak
paper lens
re-optimized
target-energy gain G
100
10
1
0.1
(b) redirection: broadband
paper lens
re-optimized
paper lens, $|R| \leq 0.043$
paper lens, $|R| \leq 0.015$
ray prediction
redirection G_φ
10
8
6
4
2
0
(c) whole-field growth: resonant
band average
paper lens
paper lens, fine sweep
re-optimized
design ω
interior field growth
30
10
3
1
0.3
0.1
20
30
40
50
60
drive frequency ω
(d) injected power absorbed in the target
paper lens
re-optimized
power routed, Π/Π_0
12
10
8
6
4
2
0
20
30
40
50
60
drive frequency ω